

Lecture #3: Data in materials science

Previously on

- What is materials informatics
 - Solving materials science problems with **data** science tools
 - HW1 announcement
- Python for materials modeling
 - Automation vs. Manual
 - Intro to ASE and Pymatgen

Agenda: From materials data to materials databases

1. What data do we generate?

Experimental and computational data

2. Where can we get data?

Literature, datasets, databases

3. Can we trust and reuse it?

Data quality and FAIR principles

4. What can we do with it?

The Materials Project and materials screening

Goals

By the end of this class, you should be able to:

- Explain the role of data in materials science
- Identify and evaluate major sources of materials data
- Understand how to access and use the Materials Project

What is data?

“Data is a representation of information stored in a systematic way for the purpose of inference, argument or decision making”

From [An Introduction to Data Analysis](#) by Michael Franke

As materials science researchers,
we generate a lot of experimental
and computational data

2-3D Images:

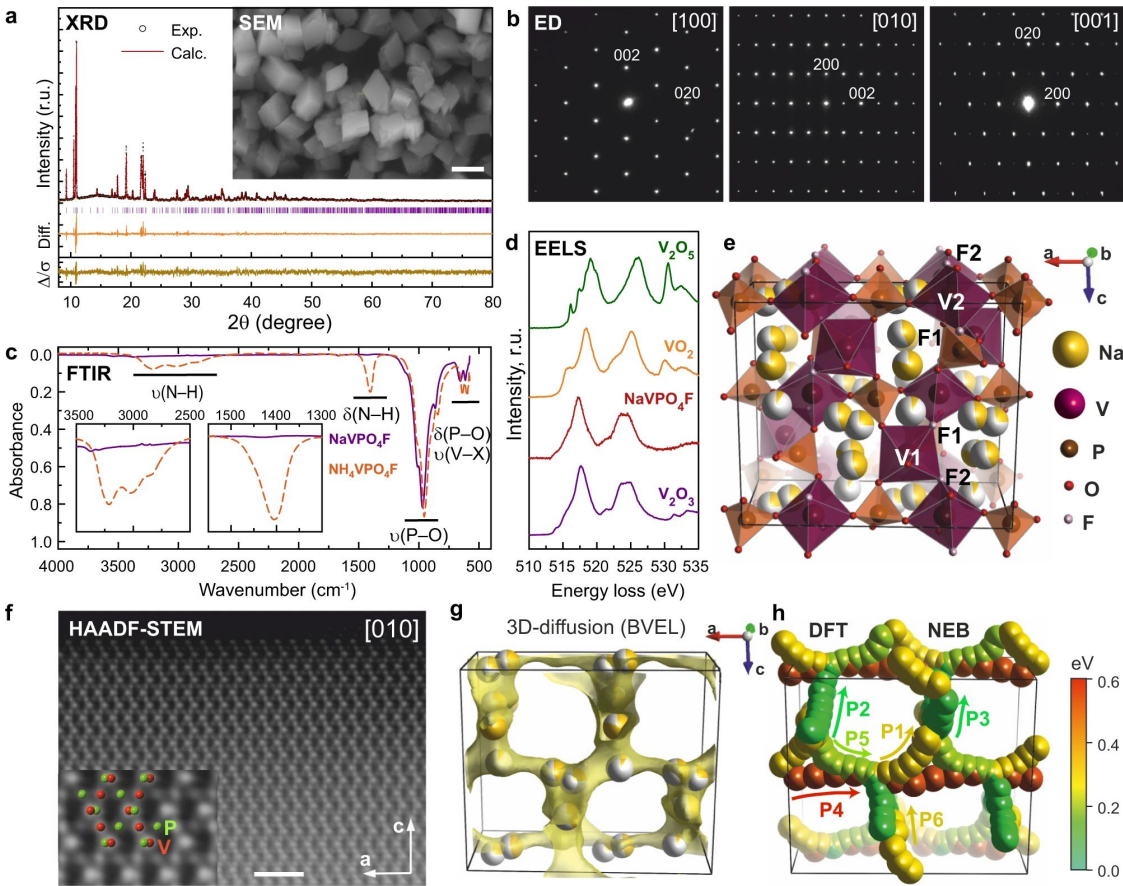
- Electron microscopy images (exp)
- Electron density (exp/comp)

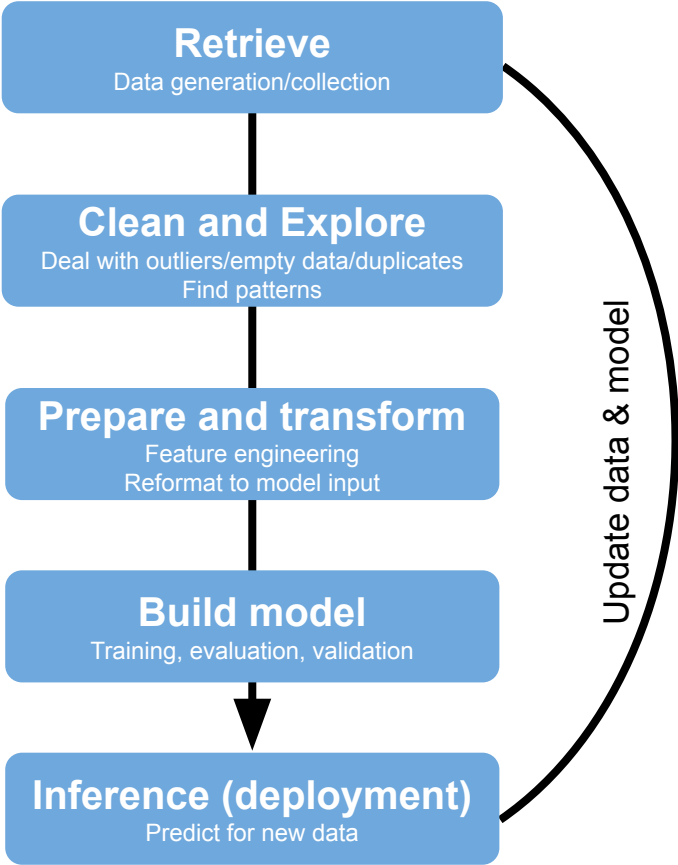
Tabular data

- E_g vs. composition (the last seminar)
- X-ray diffraction pattern (exp/comp)
- Particle size distribution (exp)
- Density of states (comp/exp)
- Energy profile (comp/exp)

Text

- Structure files (exp/comp)





Typical materials informatics workflow

Today's class

Retrieve
Data generation/collection

Where to get data?

On your own:

- Manual collection from your experiments/simulations
- High-throughput experiment or calculations
- Extract from review articles: [WebPlotDigitizer](#)
- Text mining from research articles using LLMs

From available datasets:

- Data purchase
- Open materials databases

There is a huge amount of materials science data

... scattered across various papers, datasets, databases and websites

Github:

- <https://github.com/blaiszik/Materials-Databases>

Curated databases

- ICSD: <https://icsd.products.fiz-karlsruhe.de/>
- COD: <https://www.crystallography.net/cod/>
- CCDC: <https://www.ccdc.cam.ac.uk/>
- Materials project: <https://next-gen.materialsproject.org/>
- The Open Quantum Materials Database: <http://oqmd.org/>
- AFLOW: <http://www.aflowlib.org/>
- matbench: <https://matbench.materialsproject.org/>

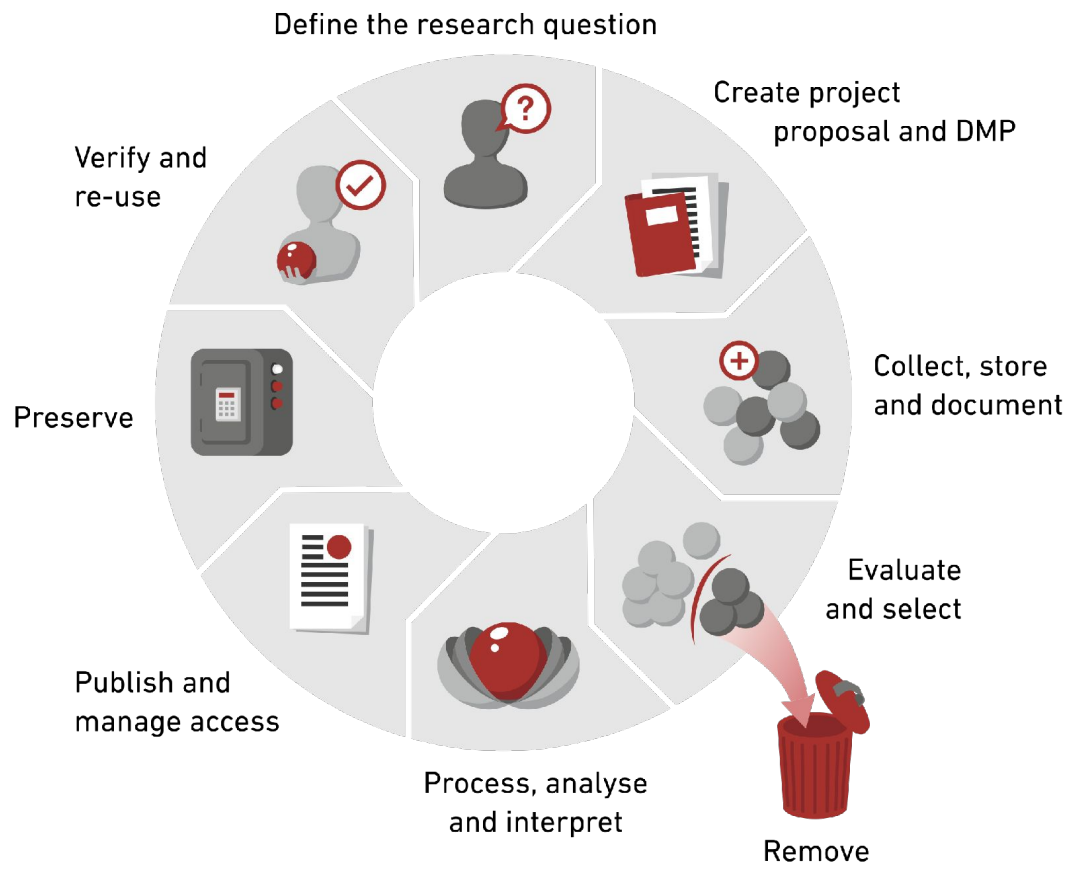
...and more.

So, before starting calculation campaign, check has somebody already generated required data?

Why use someone else's data?

Why use someone else's data?

- guide to your research objective
- reference
- baseline
- insight
- explanation
- enrichment
- time



What do we expect from the data source?

The FAIR Guiding principles

... for scientific **data management and stewardship**

- **F** is about discovery and permanent reference.
It answers "Does this data exist?"
- **A** is about the retrieval process.
It answers "What are the rules and route for getting the data?"
- **I** is about speaking a common language.
It answers "Can I merge this data with another dataset without a huge manual cleanup effort?"
- **R** is about trust and context for future use.
It answers "Do I have everything I need to use this data correctly and ethically in my own work?"



Findable



Accessible



Interoperable



Reusable

These principles should guide your projects (and future research)

Bad (toy) example - SuperDuperConductorsDB

- a dataset of superconductivity temperatures of 5,000,000 novel stable crystal structures
- with only transition temperatures and chemical compositions shared
- no crystal structures, no methods provided



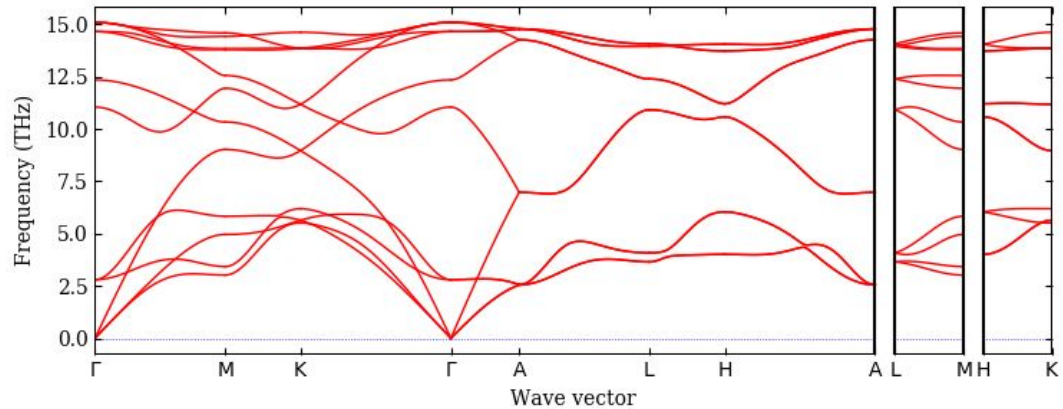
website: <https://idontsharemydata.com/>

Ok example - phonondb

- phonon dispersion curves calculated for ~10,000 crystal structures
- methods, metadata, structure files provided
- hard to handle the data

website:

<https://github.com/atztogo/phonondb>



Brilliant example - The Materials Project

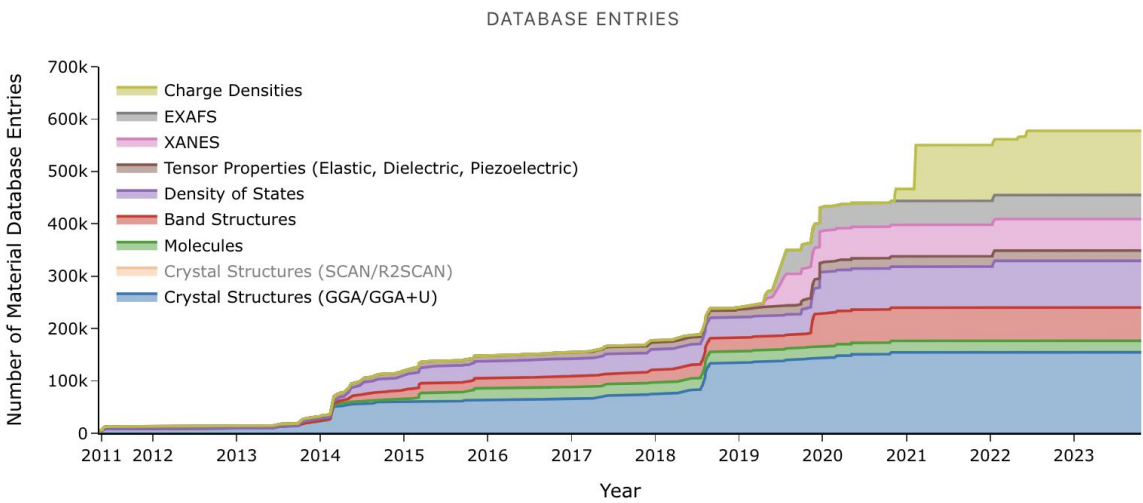
(MP) database

- ~200,000 crystal structures
- high quality density functional theory calculations
- “provides one of the largest publicly available data set of computed materials properties”
- methods, metadata, structure files provided
- handy platform
- good REST API

We will learn how to use it during today's seminar

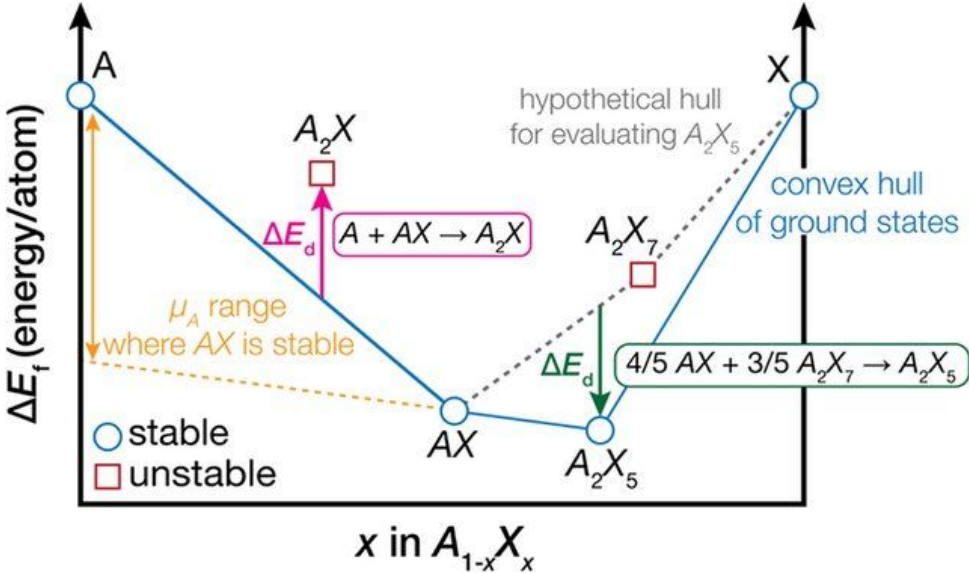
website:

<https://next-gen.materialsproject.org/>



What can we do with this data?

- Structure files
 - Input for your calculations
- Calculations results
 - Correlation analysis
 - Comparison
 - Extract more data
 - Fit a surrogate model (seminar #5)
 - Screening
 - Thermodynamics
 - Formation energies
 - Phase diagrams
 - Stability



The MP data access - simple

- you need your API key - register at materialsproject.org
- and python

Each material in the database has a unique identifier (mp_id).

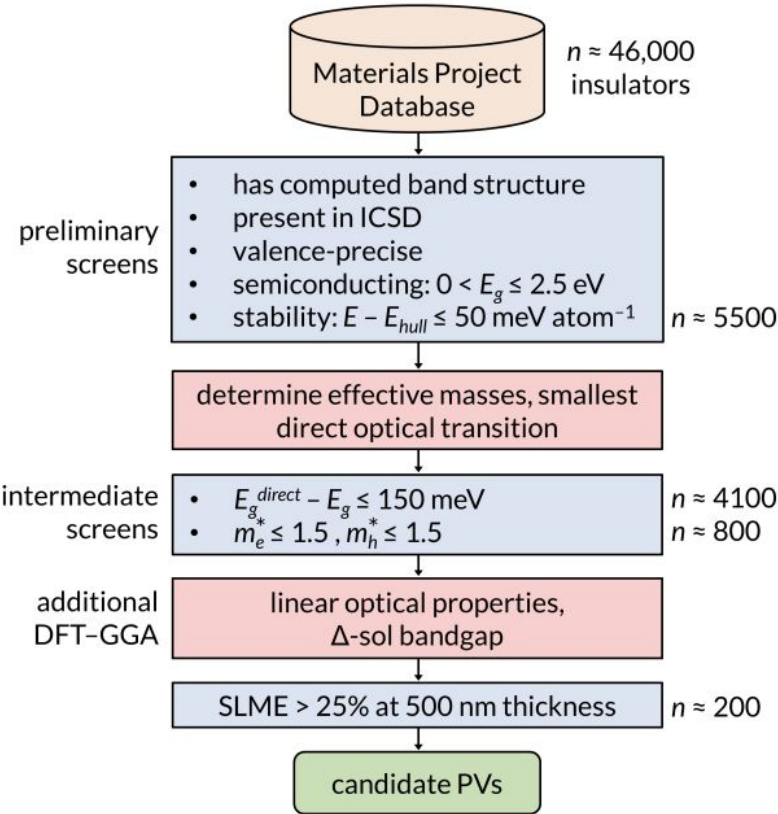
```
from mp_api.client import MPRester

with MPRester(api = "your_key") as mpr:
    # retrieve SummaryDocs for a list of materials
    docs = mpr.summary.search(material_ids=["mp-149", "mp-13"])
```

The MP usage example

- Screened database
- Identified candidates
- Calculated properties of interest for ~800 compounds
- Shared the data

Search for solar cell materials:



Spectroscopic Limited Maximum Efficiency (SLME)

Take home message

- There are many (open) sources of materials science related data
- Using someone else's data can help guide or improve your research
- Consider the FAIR principles for sharing your own data

Resources

<https://github.com/sedaoturak/data-resources-for-materials-science?tab=readme-ov-file>

https://github.com/sp8rks/MaterialsInformatics/blob/main/course_notes/5.%20Materials%20Data%20Repositories.pdf

<https://matbench.materialsproject.org/>

A few words about Density Functional Theory (DFT) before the MP database seminar

- Quantum-Mechanical approach to calculate the electronic structure (ground state electron density) of materials
- The most popular method in materials modelling
- The main (most important) data generator in computational materials science
- Has high predictive power, but:
 - scales as $O(N^3)$, N - number of electrons in a system
 - requires a lot of compute
- Out of the scope of this course
- Consider Computational Chemistry and Materials Modeling course (MA060008) for learning fundamentals of DFT